

For DATASET-NAME, the clinically appropriate causal level is **latent acute physiology and organ dysfunction**, not individual charted values. The 48-hour measurements are noisy, irregularly sampled observations of ICU physiology: blood pressure, gases, labs, urine output, neurologic scores, and ventilation status are better treated as **manifestations of hidden clinical states** than as the primary causal variables.

The strongest high-level graph is: **background/case-mix** → **chronic vulnerability and acute latent physiologic stress** → **organ dysfunction and global severity** → **observed measurements, care intensity, measurement intensity, and mortality**. This matches the dataset design: adult ICU stays, first 48 hours of irregular clinical observations, and in-hospital mortality as the prediction outcome. The official description states that DATASET-NAME contains 12,000 ICU stays, adult patients from cardiac, medical, surgical, and trauma ICUs, first-48-hour observations, six admission descriptors, and up to 37 time-varying variables.

The main identifiability limitation is that DATASET-NAME lacks diagnoses, medications, vasopressors, antibiotics, procedures, microbiology, comorbidity codes, admission diagnosis, and clinician intent. Therefore, latent states such as sepsis, shock, and cardiac injury can be approximated only as **proxy labels**, not confirmed clinical diagnoses. Missingness is also informative: EHR measurements are generated by care, values are measured at irregular intervals, and missingness may depend on expected abnormality or clinician suspicion rather than being random. JMIR Medical Inf...

1. Dataset audit

1.1 Collection design and global rules

Item	Dataset fact	Causal relevance
Population	Adult ICU stays from cardiac, medical, surgical, and trauma ICUs	Strong case-mix confounding; ICU type is not just a label but a proxy for disease spectrum and care pathway
Observation window	First 48 hours after ICU admission	Latent labels should represent first-48h physiology unless explicitly restricted to early windows
Records	12,000 ICU stays; set A has 4,000 training records, other sets were test/validation	Outcome labels are not universally available
Inclusion rule	ICU stays under 48 hours excluded	Selection/collider risk: early death or rapid discharge cases are structurally absent
Missing coding	Valid values are non-negative; -1 means missing or unknown	Must convert -1 to missing before feature extraction
Sampling	Time-stamped measurements, regular or irregular; variables may be observed once, repeatedly, or not at all	Measurement frequency is a separate observation process
Outliers	Official notes mention occasional outliers, especially with multiple sensors/methods	Robust summaries needed: first, worst, median, count, and physiologic filters

The official description specifies the first-48-hour window, six admission descriptors, 37 time-series variables, non-negative valid values, -1 missingness, irregular measurement, and occasional outliers. physionet.org

1.2 Background/static variables

Variable	Type	Clinical meaning	Unit/coding	Notes
RecordID	Irrelevant-noisy / identifier	Unique ICU stay ID	Integer	Do not use as predictor or causal variable
Age	Background	Age-related frailty and mortality vulnerability	Years	Strong confounder for mortality and severity
Gender	Background	Sex/gender marker	0 female, 1 male	Possible demographic confounder; weak physiologic specificity
Height	Background / measurement proxy	Body size	cm	Used with weight for BMI; missing often
Weight admission	Background / baseline body size	Admission weight	kg	Also appears as time-varying variable; baseline weight differs from fluid-balance weight
ICUType	Background / care-unit case mix	ICU service type	1 CCU, 2 CSRU, 3 MICU, 4 SICU	Strong proxy for admission reason, care pathway, and monitoring

1.3 Time-varying vitals, respiratory variables, and charted care variables

Variable	Type	Clinical meaning	Unit/coding	Notes
HR	Observed time-varying	Heart rate	bpm	Proxy for shock, inflammation, cardiac strain, global severity
RespRate	Observed time-varying	Respiratory rate	breaths/min	Proxy for respiratory failure, acidosis, inflammatory stress
Temp	Observed time-varying	Temperature	°C	Proxy for inflammation, infection, thermoregulatory failure
SysABP	Observed time-varying	Invasive systolic arterial pressure	mmHg	Proxy for circulatory state; invasive line implies monitoring intensity
DiasABP	Observed time-varying	Invasive diastolic arterial pressure	mmHg	Proxy for circulatory state
MAP	Observed time-varying	Invasive mean arterial pressure	mmHg	Key shock/hemodynamic proxy
NISysABP	Observed time-varying	Non-invasive systolic pressure	mmHg	Same physiologic construct as systolic BP, different measurement process
NIDiasABP	Observed time-varying	Non-invasive diastolic pressure	mmHg	Same physiologic construct as diastolic BP, different measurement process
NIMAP	Observed time-varying	Non-invasive mean arterial pressure	mmHg	Hemodynamic proxy; less invasive measurement
SaO2	Observed time-varying	Oxygen saturation	%	Proxy for hypoxemia; affected by oxygen therapy
PaO2	Observed time-varying	Arterial oxygen partial pressure	mmHg	ABG-derived proxy for respiratory failure; ordering is informative
PaCO2	Observed time-varying	Arterial carbon dioxide partial pressure	mmHg	Proxy for ventilation failure and acid-base disturbance
FIO2	Care-process / observed time-varying	Fraction of inspired oxygen	0–1	Treatment setting and respiratory severity proxy
MechVent	Treatment / care-process / proxy	Mechanical ventilation status	0 false, 1 true	True intervention/care process and proxy for respiratory/global severity
GCS	Observed time-varying	Glasgow Coma Scale	3–15	Neurologic dysfunction proxy; sedation and ventilation confounding possible
Urine	Observed time-varying / charted process	Urine output	mL	Renal perfusion/function proxy; charting intensity matters
Weight time series	Observed time-varying / care-process proxy	Serial weight	kg	Often used for fluid balance; noisy and measurement-process dependent

1.4 Laboratory variables

Variable	Type	Clinical meaning	Unit/coding	Notes
Albumin	Observed lab proxy	Nutrition, inflammation, hepatic synthetic reserve, chronic illness	g/dL	Non-specific; useful for chronic vulnerability and hepatic/inflammatory burden

Variable	Type	Clinical meaning	Unit/coding	Notes
ALP	Observed lab proxy	Cholestasis/hepatobiliary injury	IU/L	Hepatic dysfunction proxy
ALT	Observed lab proxy	Hepatocellular injury	IU/L	Hepatic injury proxy
AST	Observed lab proxy	Hepatocellular or muscle injury	IU/L	Hepatic/cardiac/muscle injury proxy
Bilirubin	Observed lab proxy	Cholestasis/liver dysfunction	mg/dL	SOFA hepatic component uses bilirubin
BUN	Observed lab proxy	Renal dysfunction, catabolism, hypovolemia	mg/dL	Renal/perfusion proxy
Creatinine	Observed lab proxy	Renal filtration	mg/dL	Renal dysfunction proxy; baseline unavailable
Glucose	Observed lab proxy	Metabolic stress, diabetes, nutrition/insulin effects	mg/dL	Metabolic derangement proxy
HCO3	Observed lab proxy	Metabolic acid-base status	mmol/L	Metabolic/renal/respiratory compensation proxy
HCT	Observed lab proxy	Hematocrit, anemia/hemoconcentration	%	Hematologic status and fluid balance proxy
K	Observed lab proxy	Potassium	mEq/L	Renal/metabolic derangement proxy
Lactate	Observed lab proxy	Tissue hypoperfusion, mitochondrial stress, sepsis/shock severity	mmol/L	Strong shock/metabolic/global severity proxy; ordering is informative
Mg	Observed lab proxy	Magnesium	mmol/L	Metabolic/electrolyte proxy
Na	Observed lab proxy	Sodium	mEq/L	Fluid/electrolyte/neurologic risk proxy
pH	Observed lab proxy	Arterial acid-base status	0–14	Metabolic/respiratory derangement proxy
Platelets	Observed lab proxy	Coagulation/hematologic dysfunction	cells/nL	SOFA coagulation component uses platelets
WBC	Observed lab proxy	Leukocyte response	cells/nL	Inflammation/infection/stress proxy
Cholesterol	Observed lab proxy / weak relevance	Lipid/nutritional/chronic marker	mg/dL	Sparse/low acute causal specificity
TropI	Observed lab proxy	Troponin-I myocardial injury marker	µg/L	Cardiac injury proxy; test ordering is highly informative
TropT	Observed lab proxy	Troponin-T myocardial injury marker	µg/L	Cardiac injury proxy; assay-specific threshold uncertainty

1.5 Outcome-related descriptors

Variable	Type	Clinical meaning	Unit/coding	Notes
SAPS-I score	Outcome-related severity descriptor	ICU severity score	Score	Training outcome file; avoid for latent-label construction to prevent dataset split dependence

Variable	Type	Clinical meaning	Unit/coding	Notes
SOFA score	Outcome-related severity descriptor	Organ failure score	Score	Training outcome file; useful for validation only, not for proxy-label generation
Length of stay	Outcome/post-outcome	Hospital length of stay	Days	Do not use for latent labels; post-baseline outcome process
Survival	Outcome/post-outcome	Days from ICU admission to death, or -1 if survivor	Days	Do not use; direct outcome leakage
In-hospital death	Outcome	Mortality label	0 survivor, 1 died	Primary outcome node OUT_InHospitalMortality

SAPS-I, SOFA, length of stay, survival, and in-hospital death are stored separately as outcome-related descriptors, and only set A outcomes were available to challenge participants. physionet.org

2. Candidate latent variables

Candidate latent variable	Clinical meaning	Supporting observed variables	Dataset support	Decision	Rationale
Chronic baseline vulnerability	Frailty/comorbidity/body reserve/case-mix risk	Age, ICUType, BMI, Albumin, admission Weight	Medium-low	Keep	Important for chronic disease and comorbidities
Acute global severity	Multi-domain acute physiologic instability	BP, HR, GCS, lactate, pH, HCO3, PaO2/FiO2, creatinine, urine, bilirubin, platelets	High	Keep	Clinical severity
Circulatory shock / hemodynamic instability	Hypoperfusion and inadequate circulatory support	MAP, SysABP/NISysABP, lactate, HR, urine, pH/HCO3	High	Keep	Stratification for hemodynamic process
Respiratory failure / pulmonary dysfunction	Hypoxemia, ventilatory failure, oxygen/ventilator dependence	PaO2, FiO2, SaO2, RespRate, PaCO2, pH, MechVent	High	Keep	Stratification for respiratory dysfunction
Renal dysfunction / kidney injury	Impaired filtration or urine output	Creatinine, BUN, Urine, K, HCO3	High	Keep	Stratification for renal dysfunction
Hepatic dysfunction / liver injury	Cholestasis/hepatocellular injury/synthetic reserve	Bilirubin, AST, ALT, ALP, Albumin	Medium	Keep	Bilirubin, liver function, INR
Coagulation/hematologic dysfunction	Thrombocytopenia, anemia, marrow/stress response	Platelets, HCT, WBC	Medium	Keep	Platelet count, coagulation
Inflammation / sepsis-like burden	Systemic host response, possible infection	Temp, WBC, HR, RespRate, PaCO2, Lactate, platelets	Medium-low	Keep, rename as inflammatory/sepsis-like burden	No clear clinical significance
Neurologic dysfunction	Depressed consciousness or brain dysfunction	GCS, metabolic derangements, hypoxemia/hypercapnia	Medium	Keep	GCS and neurologic-like
Cardiac injury/strain	Myocardial injury or ischemic/cardiac strain	Tropl, TropT, HR, BP, lactate, ICUType	Medium-low	Keep as weakly measured	Troponin measurement
Metabolic/acid-base/electrolyte derangement	Acidosis/alkalosis, lactate, electrolytes, glucose	pH, HCO3, Lactate, Na, K, Mg, Glucose, PaCO2	High	Keep	Broadly used for metabolic failure

Candidate latent variable	Clinical meaning	Supporting observed variables	Dataset support	Decision	Rationale
Fluid balance / volume-status burden	Fluid overload or depletion	Weight time series, urine, Na, BP	Low	Reject / merge	Weigh fluid
Treatment/care intensity	Intensity of clinical support	MechVent, FiO2, measurement counts, ABG/lactate/troponin ordering	High as process, not physiology	Model separately as care/measurement process	No de-clin
Confirmed sepsis	Infection plus dysregulated host response	Temp, WBC, lactate, organ dysfunction	Absent for confirmation	Reject as confirmed diagnosis	No sus-tir
Vasopressor-dependent shock	Shock requiring vasopressors	MAP, lactate only	Absent	Reject / approximate as shock	Vas uni
ARDS	Acute respiratory distress syndrome	PaO2/FiO2, MechVent	Low-medium	Reject as formal ARDS	No tir
Chronic kidney disease	Baseline renal disease	Creatinine, BUN	Low	Reject as separate variable	No cre

SOFA motivates respiratory, coagulation, hepatic, cardiovascular, neurologic, and renal organ dysfunction domains; the table version uses PaO2/FiO2, platelets, bilirubin, MAP/vasopressors, GCS, creatinine, and urine output. MSD Manuals Sepsis-3 motivates separating infection/inflammation from organ dysfunction because sepsis is defined as life-threatening organ dysfunction from a dysregulated host response, but DATASET-NAME lacks infection-confirming variables. jamanetwork.com

3. Final approved latent variables

Latent variable	Clinical meaning	Why latent	Observed indicators	Suggested bi definition
LAT_CHRONIC_BASELINE_RISK	Baseline vulnerability before ICU physiology	True comorbid burden/frailty is not observed	Age, ICUType, BMI, Albumin	Score ≥2 from habitus, low a high-risk ICU
LAT_GLOBAL_SEVERITY	Multi-system acute illness severity	Severity is a latent synthesis of multiple organ states	Hemodynamics, respiratory, neuro, renal, hepatic/coag, metabolic, cardiac domains	Score ≥3 don abnormal, or critical abnor
LAT_SHOCK	Circulatory failure/hypoperfusion	Shock is not equal to any single BP or lactate value	MAP/SBP, lactate, HR, urine, pH/HCO3	Score ≥2 sho
LAT_RESPIRATORY_FAILURE	Hypoxemia/ventilatory failure or ventilator dependence	Oxygenation state depends on disease and treatment settings	PaO2/FiO2, SaO2, RespRate, PaCO2, pH, MechVent	Score ≥2 resp indicators
LAT_RENAL_DYSFUNCTION	Reduced filtration/oliguria or renal-metabolic consequences	Renal function state is not fully observed without baseline	Creatinine, BUN, Urine, K, HCO3	Score ≥2 ren
LAT_HEPATIC_DYSFUNCTION	Liver injury/cholestasis/synthetic reserve impairment	No direct liver diagnosis or INR	Bilirubin, AST, ALT, ALP, Albumin	Score ≥2 hep indicators, wit weighted stro
LAT_COAG_HEME_DYSFUNCTION	Thrombocytopenia/anemia/hematologic stress	Coagulation system not directly measured	Platelets, HCT, WBC	Weighted sco

Latent variable	Clinical meaning	Why latent	Observed indicators	Suggested bi definition
LAT_INFLAMMATION_SEPSIS_BURDEN	Systemic inflammatory or sepsis-like host response	Infection source and antibiotics unavailable	Temp, WBC, HR, RR/PaCO2, lactate, organ dysfunction proxies	Score ≥3, or score ≥2 plus lactate/organ
LAT_NEUROLOGIC_DYSFUNCTION	Depressed consciousness/brain dysfunction	GCS is observed but confounded by sedation and ventilation	GCS, hypoxemia/hypercapnia, Na/glucose extremes, pH	Weighted score severe GCS w
LAT_CARDIAC_INJURY_STRAIN	Myocardial injury/ischemia/strain	True cardiac injury requires clinical context	Tropl, TropT, HR, hypotension, lactate, ICUType	Weighted score troponin abnormal weighted
LAT_METABOLIC_DERANGEMENT	Acid-base, lactate, electrolyte, glucose instability	Metabolic state is multidimensional	pH, HCO3, lactate, Na, K, Mg, glucose, PaCO2	Score ≥2 metabolic indicators

These latent variables are **proxy-label targets**, not ground-truth diagnoses. Causal representation learning explicitly motivates discovering high-level causal variables from low-level observations, rather than assuming raw observations are already the causal variables. [research-explor...](#)

4. Rule-based binary decision trees

All trees use first-48h summaries unless an implementation chooses an earlier landmark. `min`, `max`, `first`, `last`, `delta`, `count`, and `missing` refer to values after converting `-1` to missing and applying physiologic sanity filters.

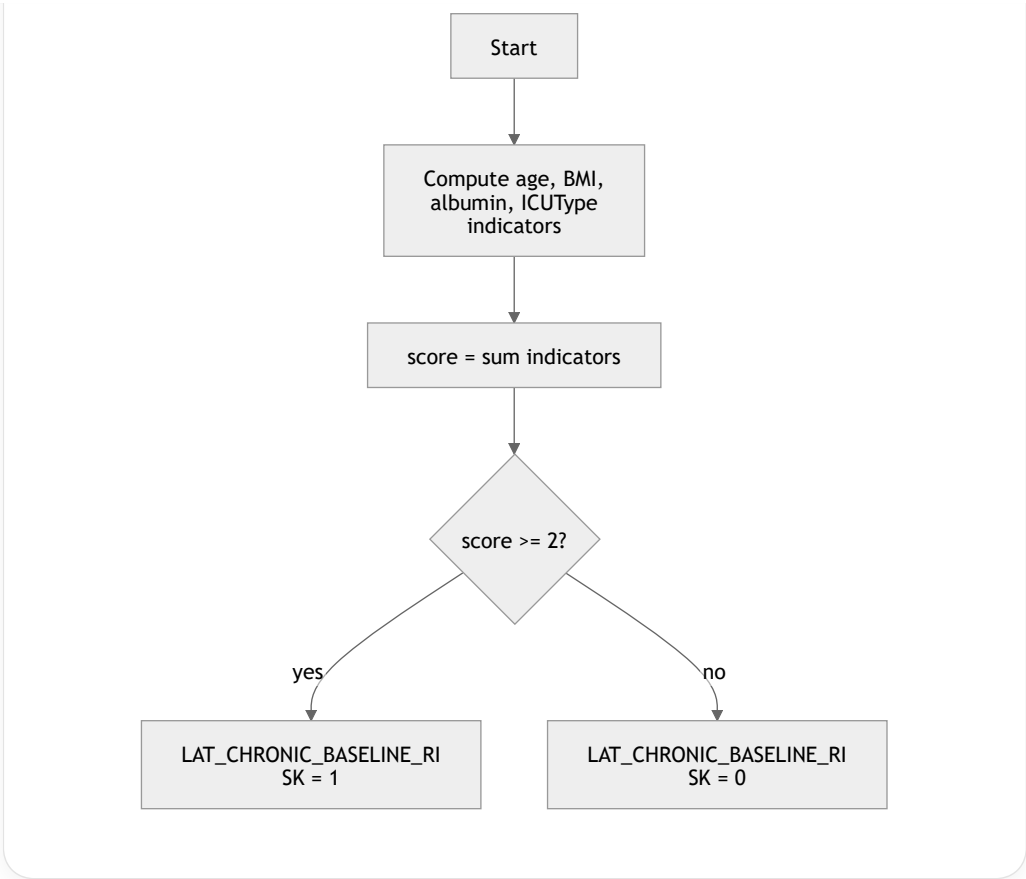
4.1 Chronic baseline vulnerability

Latent variable: LAT_CHRONIC_BASELINE_RISK

Clinical interpretation of label 1: evidence of high baseline vulnerability or limited physiologic reserve before or at ICU admission.

Component	Rule
Input variables	Age , Gender , Height , admission Weight , ICUType , Albumin
Domain indicators	AGE_RISK = Age >= 75 ; BODY_RISK = BMI < 18.5 OR BMI >= 40 ; LOW_ALBUMIN = Albumin_min < 3.0 ; CASEMIX_RISK = ICUType in {1,3}
Score	score = AGE_RISK + BODY_RISK + LOW_ALBUMIN + CASEMIX_RISK
Threshold	LAT_CHRONIC_BASELINE_RISK = 1 if score >= 2 else 0
Missingness	Missing height/weight means BODY_RISK = 0 ; missing albumin means LOW_ALBUMIN = 0 ; store available_components
Confidence	Medium-low

```
def tag_chronic_baseline_risk(x):
    age_risk = x.Age >= 75
    bmi = x.Weight_first / (x.Height / 100)**2 if x.Height and x.Weight_first else None
    body_risk = bmi is not None and (bmi < 18.5 or bmi >= 40)
    low_albumin = present(x.Albumin_min) and x.Albumin_min < 3.0
    casemix_risk = x.ICUType in [1, 3]
    score = sum([age_risk, body_risk, low_albumin, casemix_risk])
    return int(score >= 2)
```



Clinical rationale: age, poor nutritional reserve, extreme body habitus, and medical/coronary ICU case-mix can proxy chronic vulnerability.

False positives: hypoalbuminemia from acute inflammation; ICUType reflects care setting, not disease.

False negatives: younger patients with severe comorbidity not captured by the dataset.

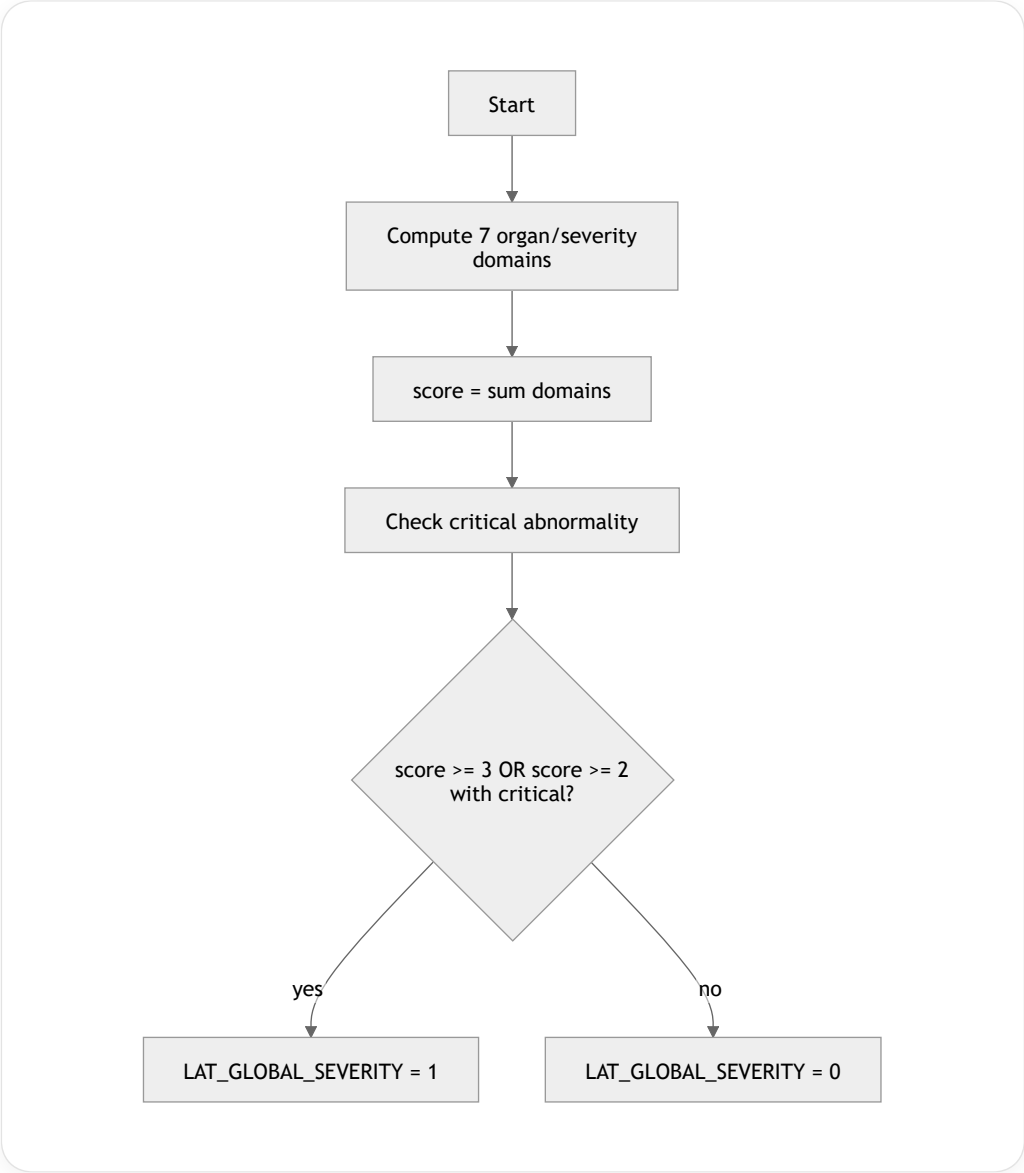
4.2 Global severity of illness

Latent variable:	LAT_GLOBAL_SEVERITY
Clinical interpretation of label 1:	evidence of multi-domain acute physiologic severity during the first 48 hours.
Component	Rule
Input variables	BP, HR, GCS, respiratory variables, renal variables, liver/coag labs, lactate, acid-base, troponin
Domain indicators	HEMO = MAP_min < 70 OR SysABP_min <= 100 OR NIMAP_min < 70 ; RESP = MechVent_max == 1 OR PF_min < 300 OR SaO2_min < 92 OR RespRate_max >= 22 ; NEURO = GCS_min < 15 ; RENAL = Creatinine_max >= 2.0 OR BUN_max >= 40 OR Urine_24h_min < 500 ; HEP_COAG = Bilirubin_max >= 2.0 OR Platelets_min < 100 ; METAB = Lactate_max > 2 OR pH_min < 7.30 OR HCO3_min < 18 ; CARDIAC = TropI_abn OR TropT_abn OR HR_max >= 130
Score	score = sum(domain indicators)
Threshold	1 if score >= 3 OR (score >= 2 AND CRITICAL == 1)
Critical	Lactate_max >= 4 OR pH_min < 7.20 OR MechVent_max == 1 OR GCS_min <= 8 OR MAP_min < 60
Missingness	Missing domains score 0; require at least two available domains for positive label unless a critical rule is present
Confidence	High

```
def tag_global_severity(x):
    hemo = any_present_true([
        lt(x.MAP_min, 70), lt(x.NIMAP_min, 70),
        le(x.SysABP_min, 100), le(x.NISysABP_min, 100)
    ])
    return hemo
```



```
resp = any_present_true([
    eq(x.MechVent_max, 1), lt(x.PF_min, 300),
    lt(x.SaO2_min, 92), ge(x.RespRate_max, 22)
])
neuro = present(x.GCS_min) and x.GCS_min < 15
renal = any_present_true([ge(x.Creatinine_max, 2.0), ge(x.BUN_max, 40), lt(x.Urine_24h_min, 500)])
hep_coag = any_present_true([ge(x.Bilirubin_max, 2.0), lt(x.Platelets_min, 100)])
metab = any_present_true([gt(x.Lactate_max, 2), lt(x.pH_min, 7.30), lt(x.HCO3_min, 18)])
cardiac = any_present_true([x.TropI_abn, x.TropT_abn, ge(x.HR_max, 130)])
score = sum([hemo, resp, neuro, renal, hep_coag, metab, cardiac])
critical = any_present_true([ge(x.Lactate_max, 4), lt(x.pH_min, 7.20), eq(x.MechVent_max, 1), le(x.GCS_min,
return int(score >= 3 or (score >= 2 and critical))
```



Clinical rationale: SOFA-like organ dysfunction is a clinically accepted severity construct using respiratory, coagulation, hepatic, cardiovascular, neurologic, and renal domains. MSD Manuals

False positives: post-operative ventilation, transient lab artifacts.

False negatives: severe disease in unmeasured domains or under-sampled patients.

4.3 Circulatory shock / hemodynamic instability

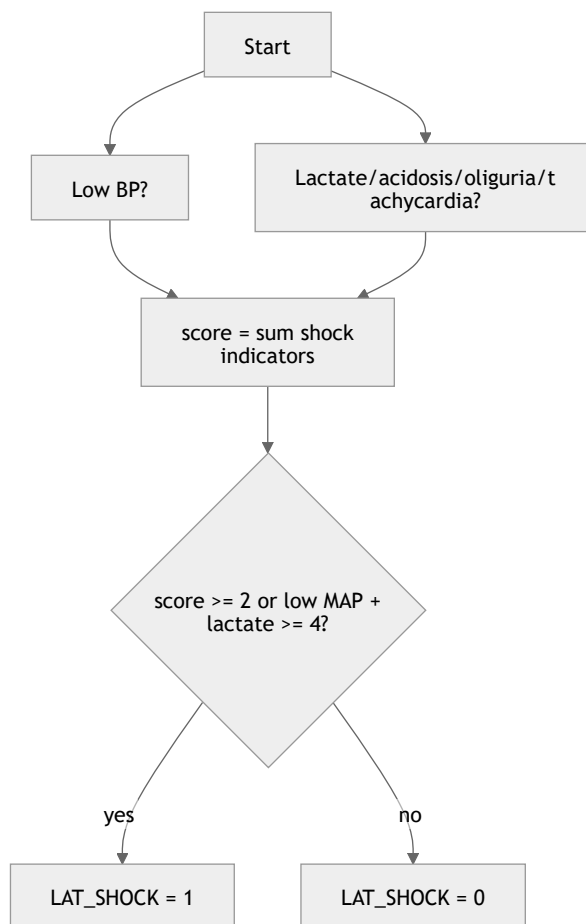
Latent variable: LAT_SHOCK

Clinical interpretation of label 1: evidence of clinically meaningful hypotension or tissue hypoperfusion.

Component	Rule
Input variables	MAP , NIMAP , SysABP , NISysABP , HR , Lactate , Urine , pH , HCO3
Domain indicators	LOW_MAP = MAP_min < 70 OR NIMAP_min < 70 ; LOW_SBP = SysABP_min <= 90 OR NISysABP_min <= 90 ; HYPOPERFUSION = Lactate_max > 2 ; TACHY = HR_max >= 110 ; OLIGURIA = Urine_24h_min < 500 ; ACIDOSIS = pH_min < 7.30 OR HCO3_min < 18

Component	Rule
Score	<code>score = sum(indicators)</code>
Threshold	1 if score ≥ 2 , OR 1 if LOW_MAP and Lactate_max ≥ 4
Missingness	Lactate missing does not rule out shock; urine missing does not imply oliguria
Confidence	High

```
def tag_shock(x):
    low_map = any_present_true([lt(x.MAP_min, 70), lt(x.NIMAP_min, 70)])
    low_sbp = any_present_true([le(x.SysABP_min, 90), le(x.NISysABP_min, 90)])
    hypoperfusion = present(x.Lactate_max) and x.Lactate_max > 2
    tachy = present(x.HR_max) and x.HR_max >= 110
    oliguria = present(x.Urine_24h_min) and x.Urine_24h_min < 500
    acidosis = any_present_true([lt(x.pH_min, 7.30), lt(x.HCO3_min, 18)])
    score = sum([low_map, low_sbp, hypoperfusion, tachy, oliguria, acidosis])
    severe_pair = low_map and present(x.Lactate_max) and x.Lactate_max >= 4
    return int(score >= 2 or severe_pair)
```



Clinical rationale: cardiovascular SOFA includes MAP <70, and lactate/acidosis/oliguria add hypoperfusion evidence. MSD Manuals

False positives: isolated low BP during sedation, measurement artifact.

False negatives: vasopressor-supported shock with normalized BP because vasopressors are unavailable.

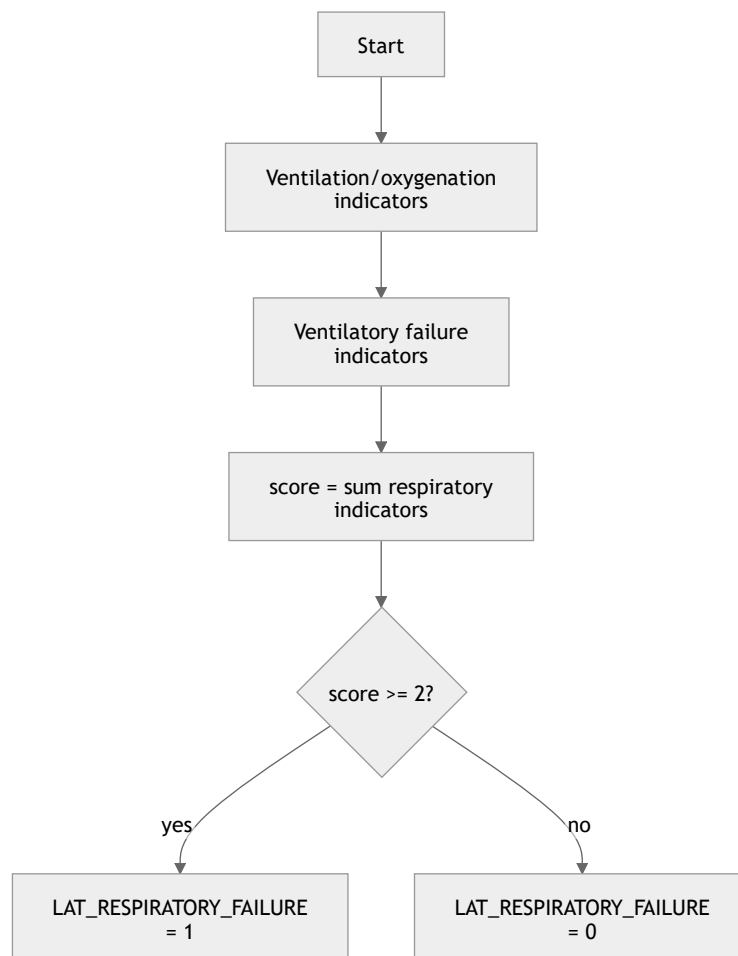
4.4 Respiratory failure / pulmonary dysfunction

Latent variable: LAT_RESPIRATORY_FAILURE

Clinical interpretation of label 1: evidence of hypoxemia, ventilatory failure, or mechanical ventilatory support.

Component	Rule
Input variables	PaO2 , FiO2 , SaO2 , RespRate , PaCO2 , pH , MechVent
Domain indicators	VENT = MechVent_max == 1 ; LOW_PF = PF_min < 300 ; HYPOXEMIA = SaO2_min < 92 OR PaO2_min < 60 ; TACHYPNEA = RespRate_max >= 22 OR RespRate_min < 8 ; VENTILATORY_FAILURE = PaCO2_max >= 50 OR (PaCO2_max >= 45 AND pH_min < 7.30)
Score	score = sum(indicators)
Threshold	1 if score >= 2 , OR 1 if VENT and LOW_PF
Missingness	Compute PaO2/FiO2 only when FiO2 and PaO2 are near-paired; missing ABG not abnormal
Confidence	High

```
def tag_respiratory_failure(x):
    vent = x.MechVent_max == 1
    low_pf = present(x.PF_min) and x.PF_min < 300
    hypoxemia = any_present_true([lt(x.SaO2_min, 92), lt(x.PaO2_min, 60)])
    tachypnea = any_present_true([ge(x.RespRate_max, 22), lt(x.RespRate_min, 8)])
    ventilatory_failure = any_present_true([
        ge(x.PaCO2_max, 50),
        (present(x.PaCO2_max) and x.PaCO2_max >= 45 and present(x.pH_min) and x.pH_min < 7.30)
    ])
    score = sum([vent, low_pf, hypoxemia, tachypnea, ventilatory_failure])
    return int(score >= 2 or (vent and low_pf))
```



Clinical rationale: PaO2/FiO2 thresholds are core SOFA respiratory markers, and mechanical ventilation is both treatment and severity proxy. MSD Manuals

False positives: elective post-operative ventilation.

False negatives: oxygen therapy without recorded FiO2 or ABG.

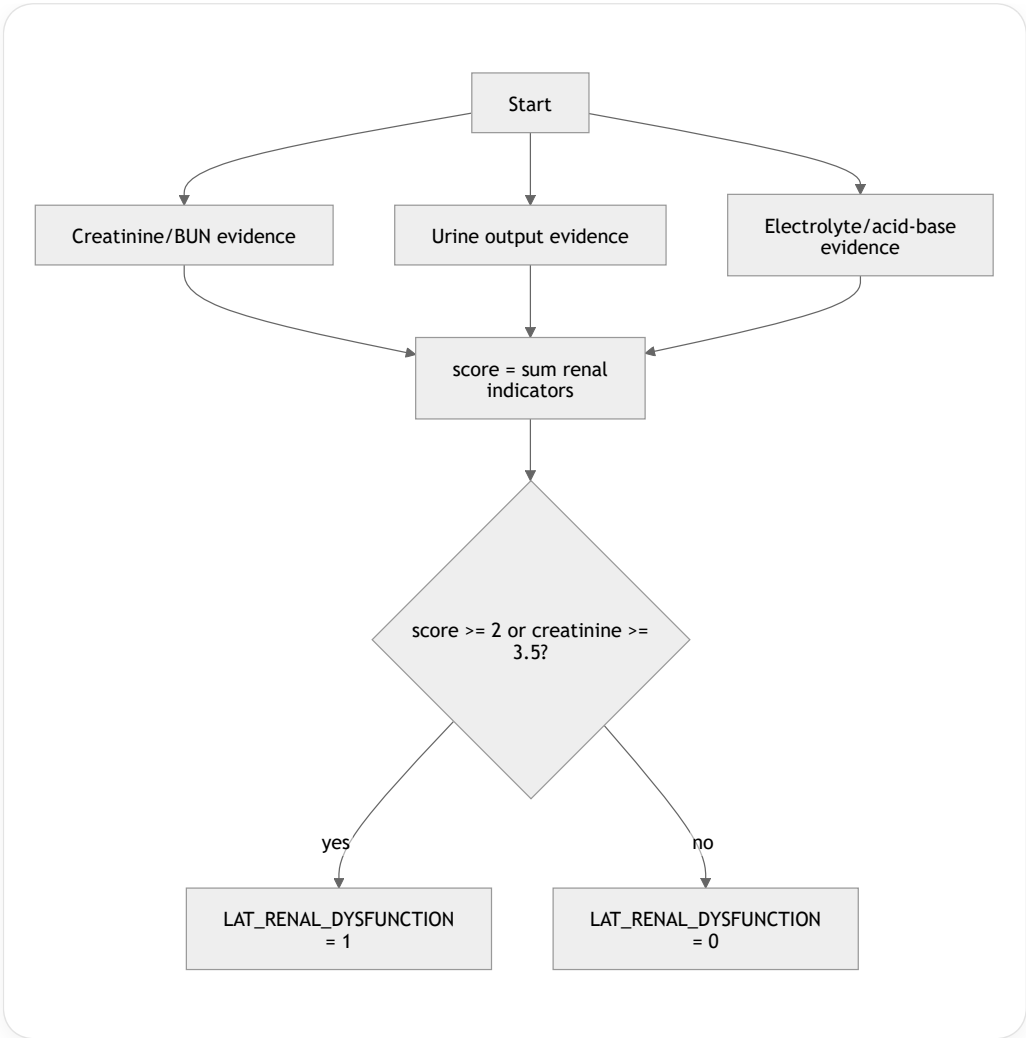
4.5 Renal dysfunction

Latent variable: LAT_RENAL_DYSFUNCTION

Clinical interpretation of label 1: evidence of renal filtration impairment, oliguria, or renal-metabolic consequences.

Component	Rule
Input variables	Creatinine , BUN , Urine , K , HCO3
Domain indicators	CREAT_HIGH = Creatinine_max >= 2.0 ; CREAT_RISE = Creatinine_max - Creatinine_first >= 0.3 ; BUN_HIGH = BUN_max >= 40 ; OLIGURIA = Urine_24h_min < 500 ; RENAL_METAB = K_max >= 5.5 OR HCO3_min < 18
Score	score = sum(indicators)
Threshold	1 if score >= 2 , Or 1 if Creatinine_max >= 3.5
Missingness	Baseline creatinine unavailable; use within-window rise only if two creatinine values exist
Confidence	High

```
def tag_renal_dysfunction(x):
    creat_high = present(x.Creatinine_max) and x.Creatinine_max >= 2.0
    creat_rise = present(x.Creatinine_first) and present(x.Creatinine_max) and (x.Creatinine_max - x.Creatinine_
    bun_high = present(x.BUN_max) and x.BUN_max >= 40
    oliguria = present(x.Urine_24h_min) and x.Urine_24h_min < 500
    renal_metab = any_present_true([ge(x.K_max, 5.5), lt(x.HCO3_min, 18)])
    score = sum([creat_high, creat_rise, bun_high, oliguria, renal_metab])
    return int(score >= 2 or ge(x.Creatinine_max, 3.5))
```



Clinical rationale: SOFA renal scoring uses creatinine and urine output, both available here. MSD Manuals
False positives: chronic kidney disease misclassified as acute dysfunction.
False negatives: early AKI without creatinine rise or uncharted urine.

4.6 Hepatic dysfunction

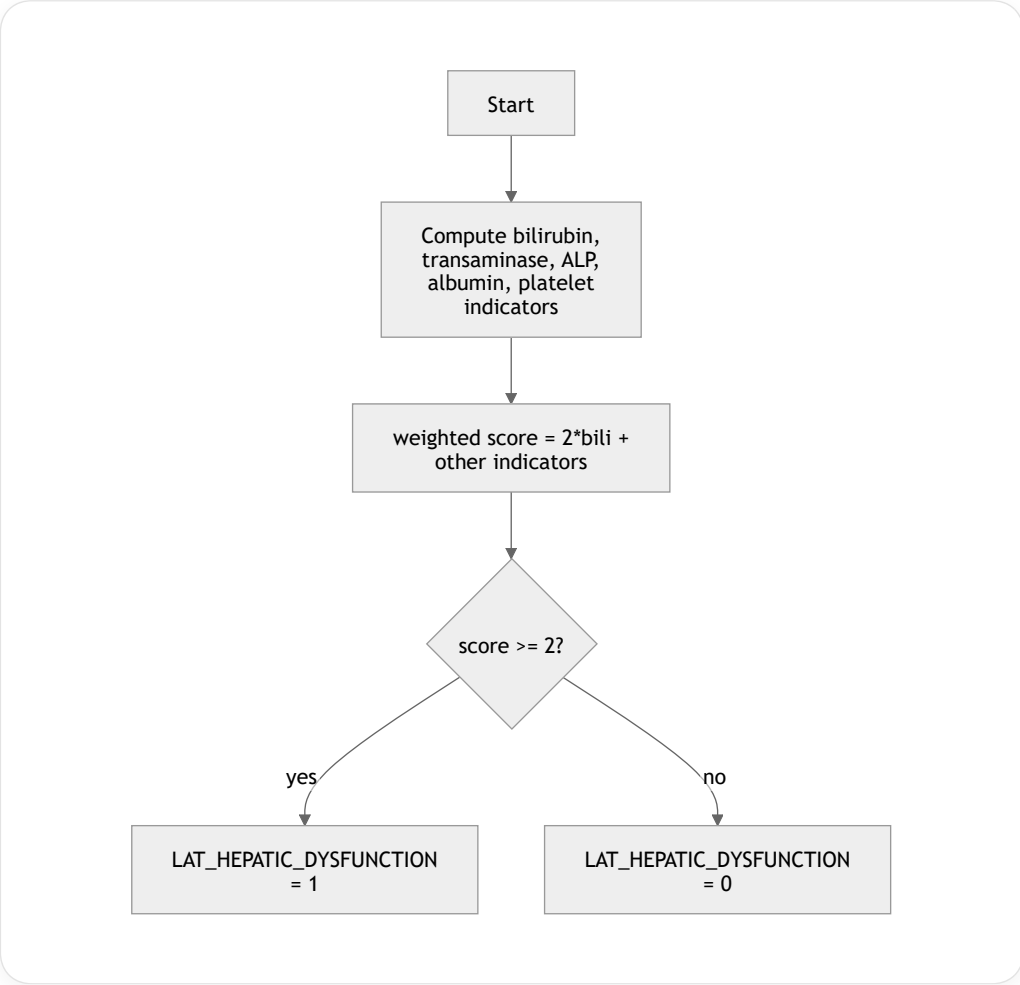
Latent variable: LAT_HEPATIC_DYSFUNCTION
Clinical interpretation of label 1: evidence of cholestatic or hepatocellular injury or poor synthetic/nutritional hepatic reserve.

Component	Rule
Input variables	Bilirubin , AST , ALT , ALP , Albumin , Platelets
Domain indicators	BILI = Bilirubin_max >= 2.0 ; TRANSAM = AST_max >= 200 OR ALT_max >= 200 ; CHOL = ALP_max >= 250 ; LOW_ALB = Albumin_min < 2.5 ; PORTAL_OR_SEVERE = Platelets_min < 100
Score	score = 2*BILI + TRANSAM + CHOL + LOW_ALB + PORTAL_OR_SEVERE
Threshold	1 if score >= 2
Missingness	Missing liver panel elements score 0 and should produce a low-evidence flag
Confidence	Medium

```
def tag_hepatic_dysfunction(x):
    bili = present(x.Bilirubin_max) and x.Bilirubin_max >= 2.0
    transam = any_present_true([ge(x.AST_max, 200), ge(x.ALT_max, 200)])
    chol = present(x.ALP_max) and x.ALP_max >= 250
    low_alb = present(x.Albumin_min) and x.Albumin_min < 2.5
    portal_or_severe = present(x.Platelets_min) and x.Platelets_min < 100
```

<> Python

Run



Clinical rationale: bilirubin is the SOFA hepatic marker; AST/ALT/ALP/albumin add injury and reserve context.

MSD Manuals

False positives: chronic liver disease, hemolysis, nutrition-related hypoalbuminemia.

False negatives: coagulopathy without bilirubin elevation because INR/PT are absent.

4.7 Coagulation or hematologic dysfunction

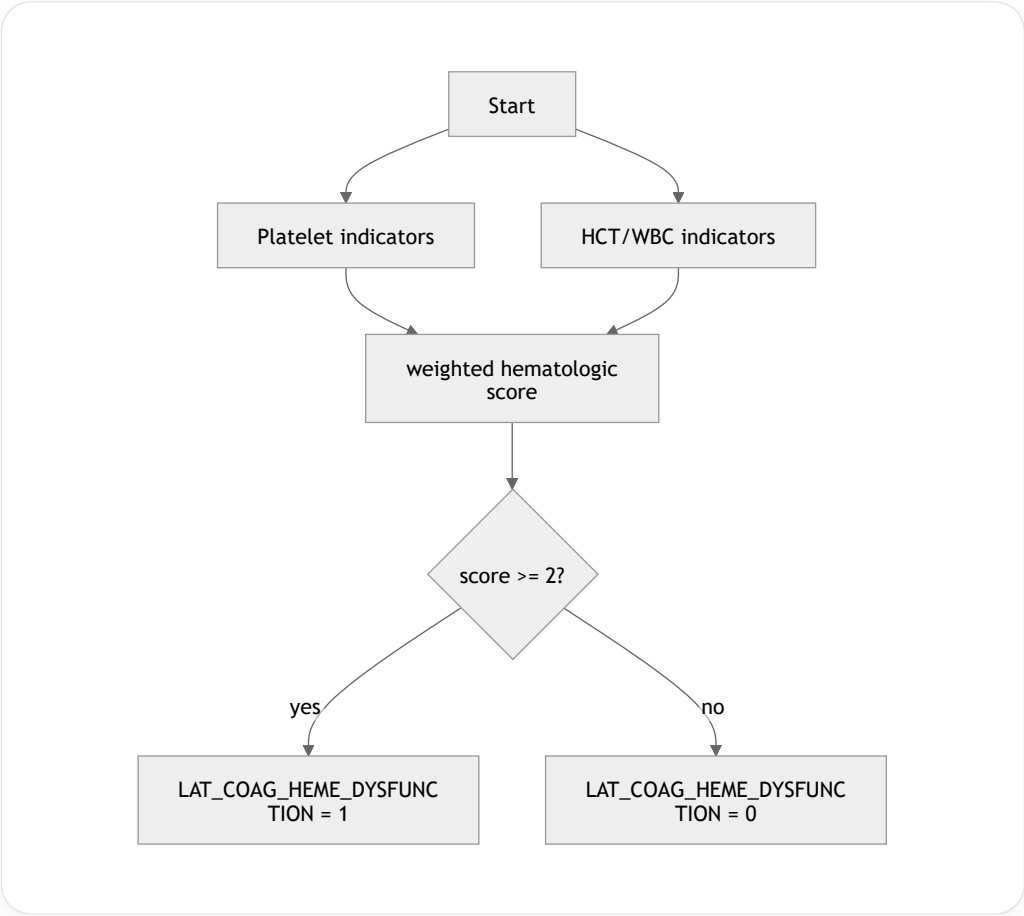
Latent variable: LAT_COAG_HEME_DYSFUNCTION

Clinical interpretation of label 1: evidence of thrombocytopenia, anemia, or hematologic stress.

Component	Rule
Input variables	Platelets , HCT , WBC
Domain indicators	SEVERE_PLT = Platelets_min < 100 ; MILD_PLT = Platelets_min < 150 ; HCT_ABN = HCT_min < 25 OR HCT_max > 55 ; WBC_EXTREME = WBC_min < 4 OR WBC_max > 20 ; PLT_DROP = Platelets_first - Platelets_min >= 50
Score	score = 2*SEVERE_PLT + MILD_PLT + HCT_ABN + WBC_EXTREME + PLT_DROP
Threshold	1 if score >= 2
Missingness	Missing CBC components score 0; do not infer normality
Confidence	Medium

```
def tag_coag_heme_dysfunction(x):
    severe_plt = present(x.Platelets_min) and x.Platelets_min < 100
    mild_plt = present(x.Platelets_min) and x.Platelets_min < 150
    hct_abn = any_present_true([lt(x.HCT_min, 25), gt(x.HCT_max, 55)])
    wbc_extreme = any_present_true([lt(x.WBC_min, 4), gt(x.WBC_max, 20)])
```

```
plt_drop = present(x.Platelets_first) and present(x.Platelets_min) and (x.Platelets_first - x.Platelets_min
score = 2*int(severe_plt) + sum([mild_plt, hct_abn, wbc_extreme, plt_drop])
return int(score >= 2)
```



Clinical rationale: SOFA coagulation uses platelet count, while HCT and WBC add hematologic context.

MSD Manuals

False positives: chronic thrombocytopenia, dilutional anemia.

False negatives: coagulopathy without thrombocytopenia because INR/PT/PTT unavailable.

4.8 Inflammatory / sepsis-like burden

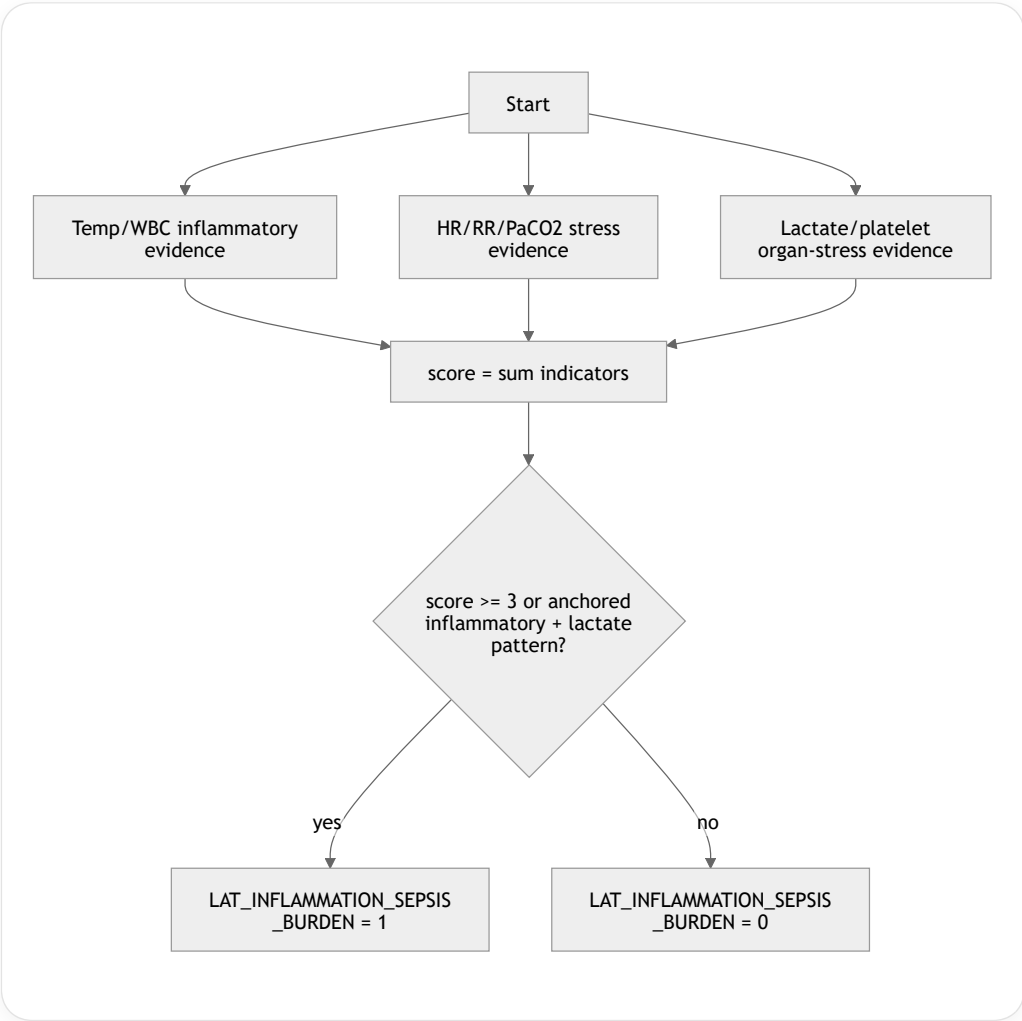
Latent variable: LAT_INFLAMMATION_SEPSIS_BURDEN

Clinical interpretation of label 1: systemic inflammatory response with possible infection-related organ stress, not confirmed sepsis.

Component	Rule
Input variables	Temp , WBC , HR , RespRate , PaCO2 , Lactate , Platelets
Domain indicators	TEMP_ABN = Temp_max > 38.3 OR Temp_min < 36 ; WBC_ABN = WBC_max > 12 OR WBC_min < 4 ; TACHY = HR_max > 90 ; RESP_STRESS = RespRate_max > 20 OR PaCO2_min < 32 ; LACTATE_STRESS = Lactate_max > 2 ; PLT_LOW = Platelets_min < 150
Score	score = sum(indicators)
Threshold	1 if score >= 3 , Or 1 if TEMP_ABN + WBC_ABN >= 1 and LACTATE_STRESS and score >= 2
Missingness	Missing WBC/temp should not be treated as absence of inflammation
Confidence	Medium-low

```
def tag_inflammation_sepsis_burden(x):
    temp_abn = any_present_true([gt(x.Temp_max, 38.3), lt(x.Temp_min, 36)])
    wbc_abn = any_present_true([gt(x.WBC_max, 12), lt(x.WBC_min, 4)])
    tachy = present(x.HR_max) and x.HR_max > 90
    resp_stress = any_present_true([gt(x.RespRate_max, 20), lt(x.PaCO2_min, 32)])
```

```
lactate_stress = present(x.Lactate_max) and x.Lactate_max > 2
plt_low = present(x.Platelets_min) and x.Platelets_min < 150
score = sum([temp_abn, wbc_abn, tachy, resp_stress, lactate_stress, plt_low])
anchored = (temp_abn or wbc_abn) and lactate_stress and score >= 2
return int(score >= 3 or anchored)
```



Clinical rationale: Sepsis-3 links dysregulated host response to organ dysfunction, but this dataset lacks infection confirmation, so the label is intentionally “sepsis-like burden.” jamanetwork.com

False positives: trauma, surgery, pancreatitis, cardiogenic shock.

False negatives: immunosuppressed or elderly patients without fever or leukocytosis.

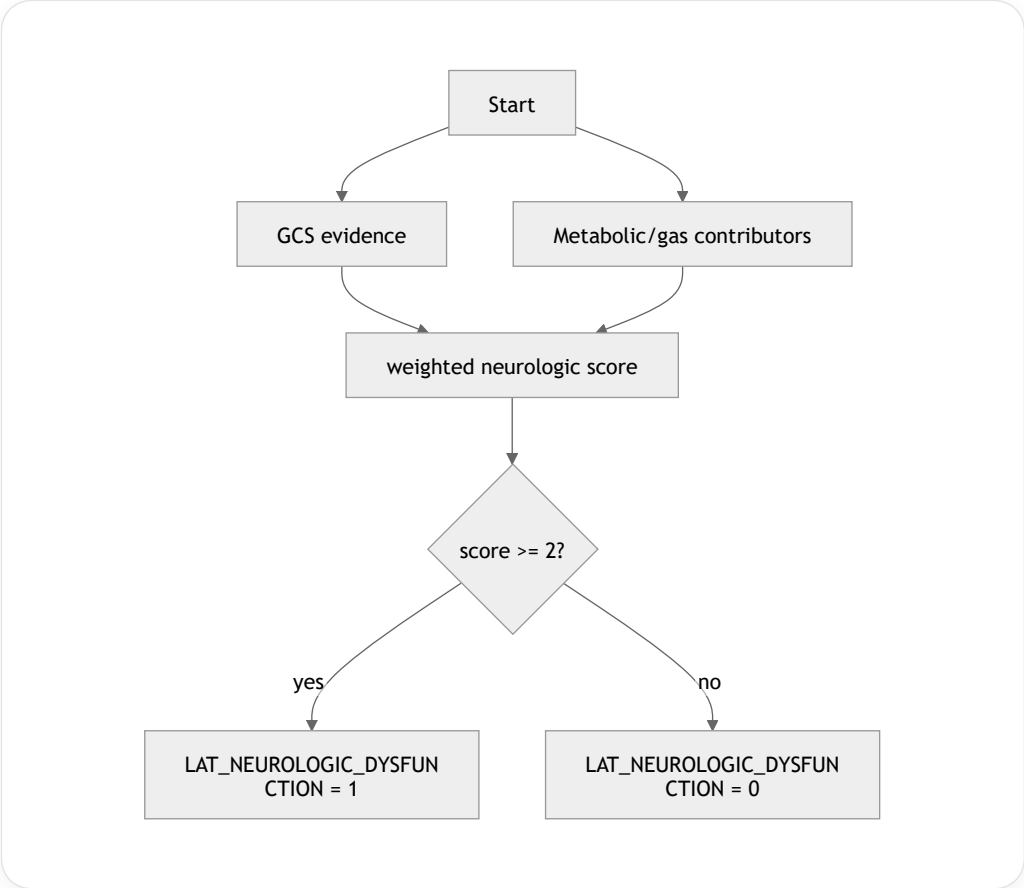
4.9 Neurologic dysfunction

Latent variable: LAT_NEUROLOGIC_DYSFUNCTION

Clinical interpretation of label 1: depressed consciousness or neurologic dysfunction, acknowledging sedation ambiguity.

Component	Rule
Input variables	GCS , Na , Glucose , PaCO2 , SaO2 , pH , MechVent
Domain indicators	GCS_MILD = GCS_min < 15 ; GCS_SEVERE = GCS_min <= 12 ; METAB_NEURO = Na_min < 130 OR Na_max > 150 OR Glucose_min < 70 OR Glucose_max > 300 ; GAS_NEURO = PaCO2_max >= 50 OR SaO2_min < 90 OR pH_min < 7.25 ; VENT_CONTEXT = MechVent_max == 1
Score	score = 2*GCS_SEVERE + GCS_MILD + METAB_NEURO + GAS_NEURO
Threshold	1 if score >= 2
Missingness	Missing GCS scores 0; ventilation is stored as ambiguity/context flag, not a score point
Confidence	Medium


```
def tag_neurologic_dysfunction(x):
    gcs_mild = present(x.GCS_min) and x.GCS_min < 15
    gcs_severe = present(x.GCS_min) and x.GCS_min <= 12
    metab_neuro = any_present_true([lt(x.Na_min, 130), gt(x.Na_max, 150), lt(x.Glucose_min, 70), gt(x.Glucose_max, 180)])
    gas_neuro = any_present_true([ge(x.PaCO2_max, 50), lt(x.SaO2_min, 90), lt(x.pH_min, 7.25)])
    score = 2*int(gcs_severe) + sum([gcs_mild, metab_neuro, gas_neuro])
    return int(score >= 2)
```



Clinical rationale: SOFA neurologic scoring uses GCS, but GCS can be affected by sedation, ventilation, and metabolic disturbance. MSD Manuals

False positives: sedated or intubated patients.

False negatives: neurologic injury without recorded GCS.

4.10 Cardiac injury or cardiac strain

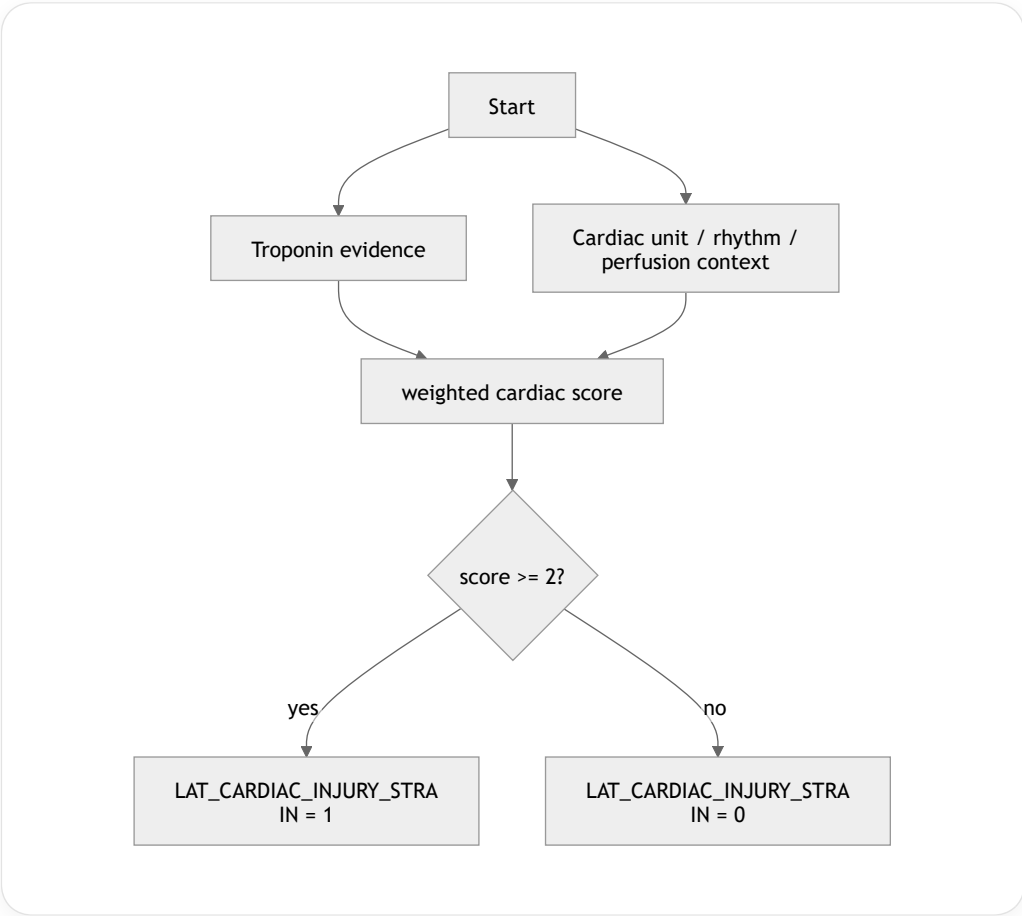
Latent variable: LAT_CARDIAC_INJURY_STRAIN

Clinical interpretation of label 1: evidence of myocardial injury, ischemia, or severe cardiac strain.

Component	Rule
Input variables	TropI , TropT , HR , MAP , SysABP , Lactate , ICUType
Domain indicators	TROP_ABN = TropI_max > 0.1 OR TropT_max > 0.01 ; CARDIAC_UNIT = ICUType in {1,2} ; ARRHYTHMIC_STRESS = HR_max >= 130 OR HR_min < 50 ; HEMO_STRAIN = MAP_min < 70 OR SysABP_min <= 90 ; PERFUSION_STRESS = Lactate_max > 2
Score	score = 2*TROP_ABN + CARDIAC_UNIT + ARRHYTHMIC_STRESS + HEMO_STRAIN + PERFUSION_STRESS
Threshold	1 if score >= 2
Missingness	Missing troponin does not imply no cardiac injury; output troponin_measured flag
Confidence	Medium-low

```
def tag_cardiac_injury_strain(x):
    trop_abn = any_present_true([gt(x.TropI_max, 0.1), gt(x.TropT_max, 0.01)])
    cardiac_unit = x.ICUType in [1, 2]
```

```
arrhythmic_stress = any_present_true([ge(x.HR_max, 130), lt(x.HR_min, 50)])
hemo_strain = any_present_true([lt(x.MAP_min, 70), le(x.SysABP_min, 90)])
perfusion_stress = present(x.Lactate_max) and x.Lactate_max > 2
score = 2*int(trop_abn) + sum([cardiac_unit, arrhythmic_stress, hemo_strain, perfusion_stress])
return int(score >= 2)
```



Clinical rationale: troponin is specific when measured, while HR, hypotension, and lactate capture strain and hypoperfusion context.

False positives: demand ischemia, renal failure-related troponin elevation, post-operative cardiac monitoring.

False negatives: unmeasured troponin or assay-specific thresholds.

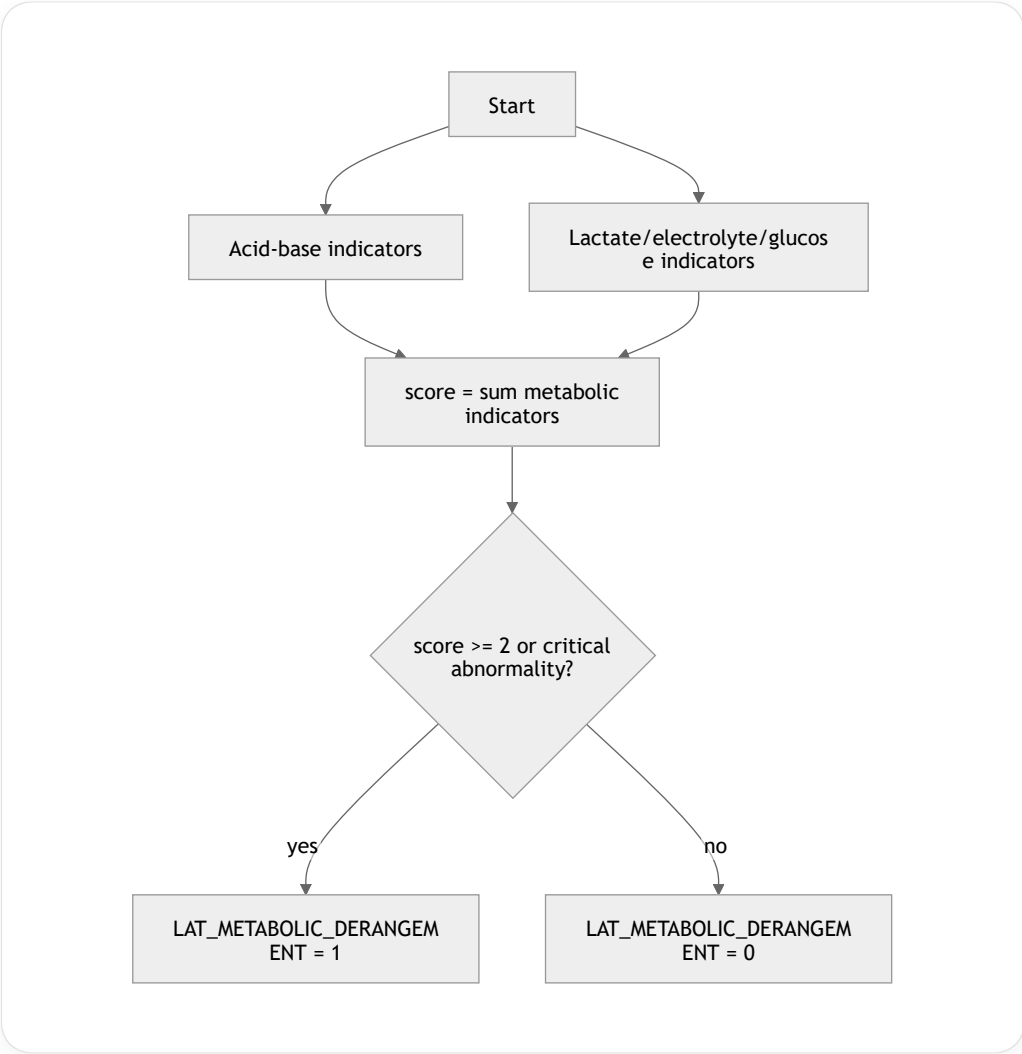
4.11 Metabolic, acid-base, and electrolyte derangement

Latent variable: LAT_METABOLIC_DERANGEMENT

Clinical interpretation of label 1: clinically meaningful acid-base, lactate, electrolyte, or glucose derangement.

Component	Rule
Input variables	pH , HCO3 , Lactate , Na , K , Mg , Glucose , PaCO2
Domain indicators	PH_ABN = pH_min < 7.30 OR pH_max > 7.50 ; BICARB_ABN = HCO3_min < 18 OR HCO3_max > 32 ; LACTATE_ABN = Lactate_max > 2 ; ELECTROLYTE_ABN = Na_min < 130 OR Na_max > 150 OR K_min < 3.0 OR K_max > 5.5 OR Mg_min < 0.6 OR Mg_max > 1.2 ; GLUCOSE_ABN = Glucose_min < 70 OR Glucose_max > 250 ; VENT_COMP = PaCO2_min < 32 OR PaCO2_max > 50
Score	score = sum(indicators)
Threshold	1 if score >= 2 , or 1 if pH_min < 7.20 OR Lactate_max >= 4 OR K_max >= 6.0
Missingness	Missing ABG/electrolytes score 0; store evidence counts
Confidence	High

```
def tag_metabolic_derangement(x):
    ph_abn = any_present_true([lt(x.pH_min, 7.30), gt(x.pH_max, 7.50)])
    bicarb_abn = any_present_true([lt(x.HCO3_min, 18), gt(x.HCO3_max, 32)])
    lactate_abn = present(x.Lactate_max) and x.Lactate_max > 2
    electrolyte_abn = any_present_true([
        lt(x.Na_min, 130), gt(x.Na_max, 150),
        lt(x.K_min, 3.0), gt(x.K_max, 5.5),
        lt(x.Mg_min, 0.6), gt(x.Mg_max, 1.2)
    ])
    glucose_abn = any_present_true([lt(x.Glucose_min, 70), gt(x.Glucose_max, 250)])
    vent_comp = any_present_true([lt(x.PaCO2_min, 32), gt(x.PaCO2_max, 50)])
    score = sum([ph_abn, bicarb_abn, lactate_abn, electrolyte_abn, glucose_abn, vent_comp])
    critical = any_present_true([lt(x.pH_min, 7.20), ge(x.Lactate_max, 4), ge(x.K_max, 6.0)])
    return int(score >= 2 or critical)
```



Clinical rationale: metabolic derangement is a common mediator from shock, renal failure, respiratory failure, and sepsis-like inflammation to mortality.

False positives: transient treatment-related electrolyte or glucose changes.

False negatives: missing ABG or labs in less-monitored patients.

5. Observed-to-latent measurement model

Observed variable	Parent latent variable(s)	Direction of abnormality	Specificity	Used in tree?	Care/measurer process influencer
Age	LAT_CHRONIC_BASELINE_RISK ; direct BG role	Higher worse	Non-specific	Yes	No
Gender	BG only	Context-dependent	Non-specific	No	No
Height	LAT_CHRONIC_BASELINE_RISK via BMI	Extreme BMI worse	Non-specific	Yes	Missingness possible

Observed variable	Parent latent variable(s)	Direction of abnormality	Specificity	Used in tree?	Care/measurer process influencer
Admission Weight	LAT_CHRONIC_BASELINE_RISK via BMI	Extreme BMI worse	Non-specific	Yes	Measurement error possible
ICUType	LAT_CHRONIC_BASELINE_RISK , LAT_CARDIAC_INJURY_STRAIN ; BG case-mix	Medical/cardiac units imply different risk	Non-specific	Yes	Yes
Albumin	Chronic risk, hepatic, inflammation	Low worse	Non-specific	Yes	Lab ordering
ALP	Hepatic dysfunction	High worse	Medium	Yes	Lab ordering
ALT	Hepatic dysfunction	High worse	Medium	Yes	Lab ordering
AST	Hepatic, cardiac/muscle injury	High worse	Low-medium	Yes	Lab ordering
Bilirubin	Hepatic dysfunction, global severity	High worse	Medium-high	Yes	Lab ordering
BUN	Renal dysfunction, shock/dehydration	High worse	Medium	Yes	Lab ordering
Cholesterol	Chronic/nutritional state	Low or high context-dependent	Low	No	Lab ordering
Creatinine	Renal dysfunction	High/rising worse	High for renal filtration	Yes	Lab ordering
DiasABP	Shock	Low worse	Medium	No direct, BP family	Invasive monitoring
SysABP	Shock/global severity	Low worse	Medium	Yes	Invasive monitoring
MAP	Shock/global severity	Low worse	High	Yes	Invasive monitoring
NIDiasABP	Shock	Low worse	Medium	No direct, BP family	Non-invasive frequency
NISysABP	Shock/global severity	Low worse	Medium	Yes	Non-invasive frequency
NIMAP	Shock/global severity	Low worse	High	Yes	Non-invasive frequency
FiO2	Respiratory failure; treatment intensity	High oxygen requirement worse	Medium	Yes via PF	Strong care-provider
GCS	Neurologic dysfunction/global severity	Low worse	Medium	Yes	Sedation/ventilation
Glucose	Metabolic derangement	Low or high worse	Medium	Yes	Treatment/nutrition
HCO3	Metabolic, renal, shock	Low/high abnormal	Medium	Yes	Lab/ABG order
HCT	Coag/heme, fluid status	Low/high abnormal	Medium-low	Yes	Transfusion/fluid effects unavailable

Observed variable	Parent latent variable(s)	Direction of abnormality	Specificity	Used in tree?	Care/measurer process influencer
HR	Shock, inflammation, cardiac strain, global severity	High or very low worse	Non-specific	Yes	Continuous monitoring
K	Renal/metabolic derangement	Low/high worse	Medium	Yes	Lab ordering/treatr
Lactate	Shock, metabolic, inflammation, global severity	High worse	High for hypoperfusion/stress	Yes	Ordering highly informative
Mg	Metabolic derangement	Low/high worse	Medium-low	Yes	Lab ordering/treatr
MechVent	Respiratory failure, global severity; care process	1 worse as severity proxy	Medium	Yes	True intervention/ca process
Na	Metabolic/neurologic derangement	Low/high worse	Medium	Yes	Lab ordering/treatr
PaCO2	Respiratory failure, metabolic compensation	Low/high abnormal	Medium	Yes	ABG ordering
PaO2	Respiratory failure	Low worse, but depends on FiO2	Medium	Yes	ABG/oxygen therapy
pH	Metabolic/respiratory derangement/global severity	Low or high worse	High	Yes	ABG ordering
Platelets	Coag/heme, inflammation, hepatic/global severity	Low worse	High for platelet domain	Yes	Lab ordering
RespRate	Respiratory failure, inflammation, metabolic acidosis	High or very low worse	Medium	Yes	Charting frequ
SaO2	Respiratory failure	Low worse	Medium	Yes	Oxygen therap
Temp	Inflammation/sepsis-like burden	High/low worse	Medium	Yes	Measurement frequency
TropI	Cardiac injury/strain	High worse	High when measured	Yes	Ordering highly informative
TropT	Cardiac injury/strain	High worse	High when measured	Yes	Ordering highly informative
Urine	Renal dysfunction, shock, care/charting	Low worse	Medium-high	Yes	Foley/charting intensity
WBC	Inflammation, coag/heme stress	Low/high worse	Medium	Yes	Lab ordering
Serial weight	Fluid balance/care process	Rapid gain/loss possibly worse	Low	No main tree	Measurement frequency

6. Latent causal DAG edge list

Source node	Target node	Edge type	Clinical mechanism	Confidence	Re: ba:
BG_Age	LAT_CHRONIC_BASELINE_RISK	background-to-latent	Older age increases frailty/comorbidity vulnerability	High	Clin

Source node	Target node	Edge type	Clinical mechanism	Confidence	Relevance
BG_HeightWeightBMI	LAT_CHRONIC_BASELINE_RISK	background-to-latent	Extreme body habitus affects physiologic reserve	Medium	Clinical
BG_ICUType	LAT_CHRONIC_BASELINE_RISK	background-to-latent	ICU unit captures case-mix and admission pathway	Medium	Data
BG_ICUType	LAT_CARDIAC_INJURY_STRAIN	background-to-latent	CCU/CSRU case-mix increases cardiac injury likelihood	Medium	Data
LAT_CHRONIC_BASELINE_RISK	LAT_GLOBAL_SEVERITY	latent-to-latent	Low reserve worsens physiologic response to acute illness	Medium	Clinical
LAT_CHRONIC_BASELINE_RISK	OUT_InHospitalMortality	latent-to-mortality	Baseline vulnerability may affect mortality independent of measured physiology	Medium	Clinical
LAT_INFLAMMATION_SEPSIS_BURDEN	LAT_GLOBAL_SEVERITY	latent-to-latent	Systemic inflammation can drive multi-organ dysfunction	Medium	Sequela, dysregulation, literature
LAT_INFLAMMATION_SEPSIS_BURDEN	LAT_SHOCK	latent-to-latent	Distributive shock from inflammatory vasodilation/capillary leak	Medium	Sequelae, physiology
LAT_INFLAMMATION_SEPSIS_BURDEN	LAT_COAG_HEME_DYSFUNCTION	latent-to-latent	Inflammation can lower platelets and cause hematologic stress	Medium	Clinical
LAT_CARDIAC_INJURY_STRAIN	LAT_SHOCK	latent-to-latent	Pump failure/ischemia can cause hypotension and hypoperfusion	Medium	Clinical
LAT_RESPIRATORY_FAILURE	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Hypercapnia/hypoxemia contributes to acidosis and lactate stress	Medium	Clinical
LAT_SHOCK	LAT_RENAL_DYSFUNCTION	latent-to-latent	Renal hypoperfusion causes oliguria and kidney injury	High	Clinical
LAT_SHOCK	LAT_HEPATIC_DYSFUNCTION	latent-to-latent	Hypoperfusion can cause ischemic/cholestatic liver injury	Medium	Clinical
LAT_SHOCK	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Hypoperfusion raises lactate and acidosis	High	Clinical
LAT_RENAL_DYSFUNCTION	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Kidney failure causes electrolyte and acid-base derangements	High	Clinical
LAT_HEPATIC_DYSFUNCTION	LAT_COAG_HEME_DYSFUNCTION	latent-to-latent	Liver disease can contribute to thrombocytopenia/coagulopathy	Low	Infrequent, web of biology
LAT_GLOBAL_SEVERITY	OUT_InHospitalMortality	latent-to-mortality	Multi-organ severity increases death risk	High	ICU Clinical literature
LAT_SHOCK	OUT_InHospitalMortality	latent-to-mortality	Hypoperfusion and circulatory collapse increase mortality	High	Clinical
LAT_RESPIRATORY_FAILURE	OUT_InHospitalMortality	latent-to-mortality	Hypoxemia/ventilatory failure can be fatal	High	Clinical
LAT_RENAL_DYSFUNCTION	OUT_InHospitalMortality	latent-to-mortality	Kidney failure is organ dysfunction and severity marker	Medium-high	SO
LAT_METABOLIC_DERANGEMENT	OUT_InHospitalMortality	latent-to-mortality	Severe acidosis/electrolyte derangement can precipitate death	High	Clinical

Source node	Target node	Edge type	Clinical mechanism	Confidence	Relevance
LAT_NEUROLOGIC_DYSFUNCTION	OUT_InHospitalMortality	latent-to-mortality	Depressed consciousness reflects severe brain/systemic dysfunction	Medium	SO
LAT_CARDIAC_INJURY_STRAIN	OUT_InHospitalMortality	latent-to-mortality	Myocardial injury/strain may cause or mark fatal physiology	Medium	Clinical
LAT_SHOCK	OBS_Hemodynamics	latent-to-observed	Shock manifests as low BP/tachycardia	High	Measurement
LAT_RESPIRATORY_FAILURE	OBS_RespiratoryGasExchange	latent-to-observed	Respiratory failure manifests as low PF ratio, hypoxemia, hypercapnia	High	Measurement
LAT_RENAL_DYSFUNCTION	OBS_RenalLabsUrine	latent-to-observed	Renal dysfunction manifests as creatinine/BUN/urine/K/HCO3 changes	High	Measurement
LAT_HEPATIC_DYSFUNCTION	OBS_LiverLabs	latent-to-observed	Liver dysfunction manifests as bilirubin/transaminase/albumin abnormalities	Medium-high	Measurement
LAT_COAG_HEME_DYSFUNCTION	OBS_CBCPlatelets	latent-to-observed	Hematologic dysfunction manifests as platelet/HCT/WBC abnormalities	High	Measurement
LAT_INFLAMMATION_SEPSIS_BURDEN	OBS_TempWBCInflam	latent-to-observed	Inflammation manifests as fever/hypothermia and WBC abnormalities	Medium	Measurement
LAT_NEUROLOGIC_DYSFUNCTION	OBS_GCS	latent-to-observed	Neurologic dysfunction manifests as lower GCS	Medium-high	Measurement
LAT_CARDIAC_INJURY_STRAIN	OBS_TroponinHR	latent-to-observed	Cardiac injury manifests as troponin and rhythm/HR abnormalities	Medium	Measurement
LAT_METABOLIC_DERANGEMENT	OBS_MetabolicLabsABG	latent-to-observed	Metabolic derangement manifests as pH/HCO3/lactate/electrolyte/glucose abnormalities	High	Measurement
LAT_RESPIRATORY_FAILURE	TRT_MechanicalVentilation	treatment/care-process	Respiratory failure triggers ventilation	High	Causation
LAT_GLOBAL_SEVERITY	TRT_MechanicalVentilation	treatment/care-process	Overall severity increases probability of ventilation	Medium	Concomitant
TRT_MechanicalVentilation	OBS_RespiratoryGasExchange	treatment-or-care-process	Ventilation and FiO2 affect measured gases/oxygenation	High	Clinical measurement
LAT_GLOBAL_SEVERITY	MISS_MeasurementIntensity	measurement-process	Sicker patients are measured more often	High	EH: missing data
LAT_SHOCK	MISS_LactateABGOrdering	measurement-process	Suspected hypoperfusion triggers lactate/ABG testing	High	Clinical workflow
LAT_CARDIAC_INJURY_STRAIN	MISS_TroponinOrdering	measurement-process	Suspected cardiac injury triggers troponin measurement	High	Clinical workflow
BG_ICUType	MISS_MeasurementIntensity	measurement-process	ICUs differ in monitoring/lab routines	Medium	Data workflow
MISS_MeasurementIntensity	OBS_AvailabilityCounts	measurement-process	Ordering/charting determines missingness and counts	High	EH

DAG-based causal inference requires making assumptions explicit; confounders close back-door paths, mediators lie on causal paths, and conditioning on colliders can open biasing paths. ora.ox.ac.uk

7. Mortality mechanism summary

The main mortality mechanism is not “low MAP causes death” or “high creatinine causes death” as isolated measurements. Rather, **latent severity states** cause both abnormal observations and mortality.

The dominant causal paths are:

- 1. BG_Age / BG_ICUType / BG_HeightWeightBMI → LAT_CHRONIC_BASELINE_RISK → LAT_GLOBAL_SEVERITY → OUT_InHospitalMortality .
- 2. LAT_INFLAMMATION_SEPSIS_BURDEN → LAT_SHOCK → LAT_RENAL_DYSFUNCTION → LAT_METABOLIC_DERANGEMENT → OUT_InHospitalMortality .
- 3. LAT_RESPIRATORY_FAILURE → TRT_MechanicalVentilation → OBS_RespiratoryGasExchange , while LAT_RESPIRATORY_FAILURE → LAT_METABOLIC_DERANGEMENT → OUT_InHospitalMortality .
- 4. LAT_CARDIAC_INJURY_STRAIN → LAT_SHOCK → LAT_GLOBAL_SEVERITY → OUT_InHospitalMortality .
- 5. LAT_SHOCK → OBS_MAP/SBP/Lactate/Urine/pH and LAT_SHOCK → OUT_InHospitalMortality ; the observations are downstream manifestations, not the underlying causal state itself.
- 6. LAT_GLOBAL_SEVERITY → MISS_MeasurementIntensity → OBS_AvailabilityCounts ; the act of being measured more often is a consequence of illness and clinical concern, not a randomly assigned exposure.

8. Missingness and irregular sampling analysis

DATASET-NAME should not be treated as MCAR. The official description states that variables may be observed once, repeatedly, or not at all, with irregular measurement intervals and occasional outliers.

physionet.org EHR missingness can arise because measurements are generated by clinical care, and measurement frequency can depend on how abnormal a value is expected to be; missingness can therefore be informative and may be MNAR. JMIR Medical Inf...

Variable family	Likely missingness mechanism	Why	Modeling recommendation
Routine vitals: HR, BP, Temp, RespRate	MAR / care-intensity-driven	More frequent in sicker or monitored patients	Use values plus counts/frequency
Invasive BP: SysABP , DiasABP , MAP	MNAR / care-process	Arterial lines placed in unstable patients	Separate invasive-monitoring indicator
ABGs: PaO2 , PaCO2 , pH , SaO2 , FiO2	MNAR / suspicion-driven	Ordered for respiratory/metabolic concern	Do not impute normal; include ABG-ordering sensitivity
Lactate	MNAR / suspicion-driven	Ordered when shock/sepsis/hypoperfusion suspected	Missing lactate is not normal; count/order flag useful
Troponin	MNAR / suspicion-driven	Ordered for suspected myocardial injury	Missing troponin should not mean no injury
Liver labs	MAR/MNAR	Ordered based on illness, medication, liver concern	Missing panel reduces confidence
Urine	Care-process / charting-dependent	Foley and hourly charting more likely in severe illness	Use only when available; include urine-count flag
GCS	MAR/MNAR	Sedation/intubation and neuro concern affect measurement	Ventilation context flag needed
Serial weight	Care-process-driven	Often related to fluid management workflow	Avoid primary latent definition

For binary trees, missingness should generally **not count as abnormal**. However, missingness indicators and measurement counts should be preserved in a separate feature table because they may proxy clinician suspicion and care intensity. Using missingness directly in severity labels risks encoding local practice patterns and collider bias; using it as a sensitivity feature is safer.

9. Backdoor/confounding implications

For causal effects on mortality, safe adjustment depends on the exposure.

Exposure question	Likely confounders	Likely mediators to avoid for total effect	Collider risks	Recommendation
Effect of TRT_MechanicalVentilation on mortality	Age, ICUType, chronic baseline risk, pre-ventilation respiratory failure/global severity	Post-ventilation gases, later global severity	Conditioning on measurement intensity or only ventilated patients	Need time-indexed de do not naively adjust f downstream oxygenat
Effect of high lactate as exposure	Chronic risk, ICUType, shock/sepsis/cardiac state	Metabolic derangement, renal dysfunction if downstream	Lactate ordering is a collider of clinician suspicion and severity	Prefer modeling LAT_ not lactate as causal exposure
Effect of renal dysfunction on mortality	Age, chronic risk, ICUType, shock/sepsis	Metabolic derangement may mediate renal effect	Lab ordering and complete cases	Adjust for pre-renal confounders, not downstream acidosis i effect desired
Effect of inflammation/sepsis-like burden on mortality	Age, chronic risk, ICUType	Shock, organ dysfunction, global severity	Measuring only patients with lactate/WBC can induce selection	Do not adjust for glob severity when estimati total inflammatory bur effect
Effect of ICUType on mortality	Age, baseline risk/case mix	Acute severity may mediate case-mix effect	Conditioning on 48h survivors/stays is selection	Interpret cautiously; IC is not randomized
Effect of global severity on mortality	Age, chronic risk, ICUType	Organ-specific downstream variables if defined later	Measurement intensity	Severity is outcome-pr useful for prediction, r always valid adjustmer

Binary latent tags are not automatically safe adjustment variables. LAT_GLOBAL_SEVERITY , LAT_SHOCK , LAT_METABOLIC_DERANGEMENT , and organ dysfunction tags are often **mediators or outcome-proximal proxy outcomes**. They may be appropriate for risk adjustment or prediction, but adjusting for them can block part of the causal pathway for treatment or exposure effects. DAG literature specifically warns against treating mediators like confounders and against conditioning on colliders. [ora.ox.ac.uk](#)

10. Graph specification

A. Adjacency list

BG_Age:
-> LAT_CHRONIC_BASELINE_RISK

BG_Gender:
-> LAT_CHRONIC_BASELINE_RISK

BG_HeightWeightBMI:
-> LAT_CHRONIC_BASELINE_RISK

BG_ICUType:
-> LAT_CHRONIC_BASELINE_RISK
-> LAT_CARDIAC_INJURY_STRAIN
-> MISS_MeasurementIntensity

LAT_CHRONIC_BASELINE_RISK:
-> LAT_GLOBAL_SEVERITY
-> OUT_InHospitalMortality

LAT_INFLAMMATION_SEPSIS_BURDEN:
-> LAT_GLOBAL_SEVERITY
-> LAT_SHOCK
-> LAT_COAG_HEME_DYSFUNCTION
-> OBS_TempWBCInflam

LAT_CARDIAC_INJURY_STRAIN:
-> LAT_SHOCK
-> OUT_InHospitalMortality

```

-> OBS_TroponinHR
-> MISS_TroponinOrdering

LAT_RESPIRATORY_FAILURE:
-> LAT_METABOLIC_DERANGEMENT
-> TRT_MechanicalVentilation
-> OUT_InHospitalMortality
-> OBS_RespiratoryGasExchange

LAT_SHOCK:
-> LAT_RENAL_DYSFUNCTION
-> LAT_HEPATIC_DYSFUNCTION
-> LAT_METABOLIC_DERANGEMENT
-> OUT_InHospitalMortality
-> OBS_Hemodynamics
-> MISS_LactateABGOrdering

LAT_RENAL_DYSFUNCTION:
-> LAT_METABOLIC_DERANGEMENT
-> OUT_InHospitalMortality
-> OBS_RenalLabsUrine

LAT_HEPATIC_DYSFUNCTION:
-> LAT_COAG_HEME_DYSFUNCTION
-> OBS_LiverLabs

LAT_COAG_HEME_DYSFUNCTION:
-> OUT_InHospitalMortality
-> OBS_CBCPlatelets

LAT_NEUROLOGIC_DYSFUNCTION:
-> OUT_InHospitalMortality
-> OBS_GCS

LAT_METABOLIC_DERANGEMENT:
-> OUT_InHospitalMortality
-> OBS_MetabolicLabsABG

LAT_GLOBAL_SEVERITY:
-> OUT_InHospitalMortality
-> TRT_MechanicalVentilation
-> MISS_MeasurementIntensity

TRT_MechanicalVentilation:
-> OBS_RespiratoryGasExchange

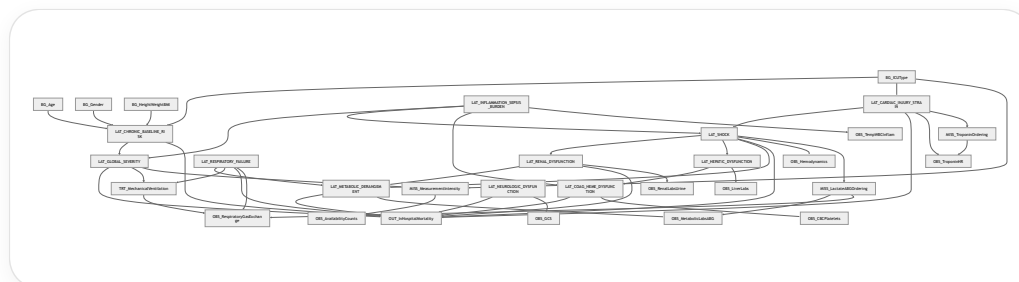
MISS_MeasurementIntensity:
-> OBS_AvailabilityCounts

MISS_LactateABGOrdering:
-> OBS_MetabolicLabsABG
-> OBS_RespiratoryGasExchange

MISS_TroponinOrdering:
-> OBS_TroponinHR

```

B. Mermaid graph



C. Python NetworkX edge list

```

edges = [
    ("BG_Age", "LAT_CHRONIC_BASELINE_RISK"),
    ("BG_Gender", "LAT_CHRONIC_BASELINE_RISK"),

```

```

("BG_HeightWeightBMI", "LAT_CHRONIC_BASELINE_RISK"),
("BG_ICUType", "LAT_CHRONIC_BASELINE_RISK"),
("BG_ICUType", "LAT_CARDIAC_INJURY_STRAIN"),
("BG_ICUType", "MISS_MeasurementIntensity"),

("LAT_CHRONIC_BASELINE_RISK", "LAT_GLOBAL_SEVERITY"),
("LAT_CHRONIC_BASELINE_RISK", "OUT_InHospitalMortality"),

("LAT_INFLAMMATION_SEPSIS_BURDEN", "LAT_GLOBAL_SEVERITY"),
("LAT_INFLAMMATION_SEPSIS_BURDEN", "LAT_SHOCK"),
("LAT_INFLAMMATION_SEPSIS_BURDEN", "LAT_COAG_HEME_DYSFUNCTION"),
("LAT_INFLAMMATION_SEPSIS_BURDEN", "OBS_TempWBCInflam"),

("LAT_CARDIAC_INJURY_STRAIN", "LAT_SHOCK"),
("LAT_CARDIAC_INJURY_STRAIN", "OUT_InHospitalMortality"),
("LAT_CARDIAC_INJURY_STRAIN", "OBS_TroponinHR"),
("LAT_CARDIAC_INJURY_STRAIN", "MISS_TroponinOrdering"),

("LAT_RESPIRATORY_FAILURE", "LAT_METABOLIC_DERANGEMENT"),
("LAT_RESPIRATORY_FAILURE", "TRT_MechanicalVentilation"),
("LAT_RESPIRATORY_FAILURE", "OUT_InHospitalMortality"),
("LAT_RESPIRATORY_FAILURE", "OBS_RespiratoryGasExchange"),

("LAT_SHOCK", "LAT_RENAL_DYSFUNCTION"),
("LAT_SHOCK", "LAT_HEPATIC_DYSFUNCTION"),
("LAT_SHOCK", "LAT_METABOLIC_DERANGEMENT"),
("LAT_SHOCK", "OUT_InHospitalMortality"),
("LAT_SHOCK", "OBS_Hemodynamics"),
("LAT_SHOCK", "MISS_LactateABGOrdering"),

("LAT_RENAL_DYSFUNCTION", "LAT_METABOLIC_DERANGEMENT"),
("LAT_RENAL_DYSFUNCTION", "OUT_InHospitalMortality"),
("LAT_RENAL_DYSFUNCTION", "OBS_RenalLabsUrine"),

("LAT_HEPATIC_DYSFUNCTION", "LAT_COAG_HEME_DYSFUNCTION"),
("LAT_HEPATIC_DYSFUNCTION", "OBS_LiverLabs"),

("LAT_COAG_HEME_DYSFUNCTION", "OUT_InHospitalMortality"),
("LAT_COAG_HEME_DYSFUNCTION", "OBS_CBCPlatelets"),

("LAT_NEUROLOGIC_DYSFUNCTION", "OUT_InHospitalMortality"),
("LAT_NEUROLOGIC_DYSFUNCTION", "OBS_GCS"),

("LAT_METABOLIC_DERANGEMENT", "OUT_InHospitalMortality"),
("LAT_METABOLIC_DERANGEMENT", "OBS_MetabolicLabsABG"),

("LAT_GLOBAL_SEVERITY", "OUT_InHospitalMortality"),
("LAT_GLOBAL_SEVERITY", "TRT_MechanicalVentilation"),
("LAT_GLOBAL_SEVERITY", "MISS_MeasurementIntensity"),

("TRT_MechanicalVentilation", "OBS_RespiratoryGasExchange"),
("MISS_MeasurementIntensity", "OBS_AvailabilityCounts"),
("MISS_LactateABGOrdering", "OBS_MetabolicLabsABG"),
("MISS_LactateABGOrdering", "OBS_RespiratoryGasExchange"),
("MISS_TroponinOrdering", "OBS_TroponinHR"),
]

import networkx as nx
G = nx.DiGraph()
G.add_edges_from(edges)
assert nx.is_directed_acyclic_graph(G)

```

11. Decision-tree implementation specification

11.1 Required input columns

RecordID, Age, Gender, Height, ICUType, Weight,
 Albumin, ALP, ALT, AST, Bilirubin, BUN, Cholesterol, Creatinine,
 DiasABP, FiO2, GCS, Glucose, HCO3, HCT, HR, K, Lactate, Mg,
 MAP, MechVent, Na, NIDiasABP, NIMAP, NISysABP, PaCO2, PaO2,
 pH, Platelets, RespRate, SaO2, SysABP, Temp, TropI, TropT,
 Urine, WBC



Do **not** use SAPS-I , SOFA , Length_of_stay , Survival , Or In-hospital_death for latent tag generation.

11.2 Derived summary features

For each time-varying variable v :

```
V_first
V_last
V_min
V_max
V_mean
V_median
V_count
V_missing = 1 if V_count == 0 else 0
V_delta = V_last - V_first when count >= 2
V_slope = robust slope over time when count >= 2
V_variability = IQR or standard deviation when count >= 2
```

Special derived features:

```
BMI = Weight_first / (Height / 100)^2
PF_ratio = PaO2 / FiO2 for near-paired measurements with FiO2 in (0, 1]
PF_min = minimum valid PF_ratio
Urine_24h_sum = sum urine output in each 24h block
Urine_24h_min = minimum available 24h urine sum
total_measurement_count = sum(V_count for all time-varying variables)
abg_count = count(PaO2, PaCO2, pH measurements)
lactate_measured = Lactate_count > 0
troponin_measured = TropI_count > 0 or TropT_count > 0
```

11.3 Expected binary output columns

```
LAT_CHRONIC_BASELINE_RISK
LAT_GLOBAL_SEVERITY
LAT_SHOCK
LAT_RESPIRATORY_FAILURE
LAT_RENAL_DYSFUNCTION
LAT_HEPATIC_DYSFUNCTION
LAT_COAG_HEME_DYSFUNCTION
LAT_INFLAMMATION_SEPSIS_BURDEN
LAT_NEUROLOGIC_DYSFUNCTION
LAT_CARDIAC_INJURY_STRAIN
LAT_METABOLIC_DERANGEMENT
```

11.4 Threshold table

Domain	Thresholds
Hypotension	MAP < 70 , NIMAP < 70 , SysABP <= 90 , NISysABP <= 90 , qSOFA-like systolic <=100 for global severity
Respiratory	PF < 300 , SaO2 < 92 , PaO2 < 60 , RespRate >=22 , PaCO2 >=50
Renal	Creatinine >=2.0 , Creatinine rise >=0.3 , BUN >=40 , Urine_24h <500 , K >=5.5
Hepatic	Bilirubin >=2.0 , AST/ALT >=200 , ALP >=250 , Albumin <2.5
Coag/heme	Platelets <150 ,severe <100 , HCT <25 , WBC <4 or >20
Inflammation	Temp >38.3 or <36 , WBC >12 or <4 , HR >90 , RR >20 , PaCO2 <32
Neurologic	GCS <15 , severe GCS <=12 , very severe GCS <=8
Cardiac	TropI >0.1 , TropT >0.01 , HR >=130 or <50
Metabolic	pH <7.30 or >7.50 ,critical pH <7.20 , HCO3 <18 or >32 , Lactate >2 ,critical >=4 , Na <130 or >150 , K <3.0 or >5.5 , Glucose <70 or >250 , Mg <0.6 or >1.2

11.5 Validation checks

```
required_checks = [
    "Convert -1 to missing before all summaries.",
    "Do not use mortality, survival, length of stay, SAPS-I, or SOFA in tag functions.",
```

```
"Ensure all generated LAT_* columns are binary 0/1.",
"Store evidence counts for each LAT_* tag.",
"Store missingness/count features separately from biological tags.",
"Check units match official dataset units.",
"Flag impossible physiologic values and either winsorize or set to missing.",
"Run DAG acyclicity check after constructing graph.",
"Perform sensitivity analyses: first 24h vs first 48h, worst-value vs first-value tags, missingness excluded"
]
```

A later preprocessing analysis of DATASET-NAME emphasized that artifacts and outliers in ICU monitor data can affect mortality models and that domain-knowledge bounds or automatic outlier rejection can improve performance. [CinC](#)

12. Rejected alternatives

Rejected item	Type	Reason
Confirmed sepsis label	Latent definition	No cultures, antibiotics, infection diagnosis, or suspected-infection timestamp
Vasopressor-dependent septic shock	Latent definition	No vasopressor medication variables
ARDS	Latent definition	No chest imaging, PEEP, bilateral infiltrates, or formal diagnosis
Chronic kidney disease	Latent definition	No pre-ICU baseline creatinine or diagnosis codes
Fluid balance latent state	Latent variable	Serial weight and urine alone are insufficient; no fluid input/diuretics
Nutritional state as separate DAG node	Latent variable	Albumin and BMI are weak and confounded by acute inflammation
Raw lab-to-raw lab dense graph	DAG structure	Confuses measurement association with causal physiology
Direct Lactate -> Mortality edge	DAG edge	Lactate is a proxy for shock/metabolic stress, not the primary causal state
Direct Creatinine -> Mortality edge	DAG edge	Creatinine measures renal dysfunction; the latent dysfunction is causal
Direct GCS -> Mortality edge	DAG edge	GCS measures neurologic dysfunction and sedation context
Using SOFA outcome descriptor to define tags	Decision rule	Training-only outcome-related file; would reduce portability and risk leakage
Using Length_of_stay Or Survival in tags	Decision rule	Direct post-outcome leakage
Treating missing lactate as normal	Missingness rule	Lactate missingness may reflect clinician expectation and ordering process
Treating troponin absence as no cardiac injury	Missingness rule	Troponin is suspicion-driven
Conditioning on measurement intensity in all mortality models	Adjustment strategy	Measurement intensity is a descendant of severity and care decisions; collider risk
Adjusting for LAT_GLOBAL_SEVERITY when estimating total effect of sepsis-like burden	Adjustment strategy	Blocks mediated effect through organ dysfunction

13. Final confidence and limitations

The strongest latent variables are LAT_GLOBAL_SEVERITY , LAT_SHOCK , LAT_RESPIRATORY_FAILURE , LAT_RENAL_DYSFUNCTION , and LAT_METABOLIC_DERANGEMENT . They are supported by multiple variables and

align with established ICU organ dysfunction constructs. SOFA's organ domains provide external clinical support for respiratory, coagulation, hepatic, cardiovascular, neurologic, and renal dysfunction scoring.

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The weakest approved latent variables are `LAT_CHRONIC_BASELINE_RISK` , `LAT_CARDIAC_INJURY_STRAIN` , and `LAT_INFLAMMATION_SEPSIS_BURDEN` . Chronic vulnerability lacks diagnoses and comorbidity codes; cardiac injury depends on sparse, suspicion-driven troponin testing; and sepsis-like burden lacks infection confirmation.

The most reliable trees are the respiratory, renal, shock, metabolic, and global severity trees. The most uncertain trees are hepatic, coagulation/heme, cardiac injury, and chronic baseline risk. The most uncertain DAG edges are hepatic dysfunction to coagulation dysfunction, chronic baseline vulnerability to global severity, and inflammation/sepsis-like burden to specific organ failures, because the dataset lacks diagnoses, medications, microbiology, vasopressors, and treatment timing.

Additional data that would improve identifiability include admission diagnoses, chronic comorbidity codes, vasopressor doses, antibiotics, cultures, fluid input, diuretics, transfusions, sedation, oxygen delivery mode, ventilator settings, PEEP, baseline creatinine, INR/PT/PTT, and timestamps for treatments. Without these, this graph should be treated as a **clinically plausible causal DAG**, not a proven ground-truth DAG. It is best used for transparent causal assumptions, latent proxy construction, sensitivity analysis, and avoiding naive adjustment for downstream severity, mediators, or colliders.

       Sources

